

## Design and Synthesis of 8-bit Shift Register using Yosys

### INTRODUCTION:

In this lab we will learn about the synthesis of 8-bit shift register using Yosys. Open source tool Yosys involves compiling Verilog code that describes a register chain, where data is shifted one bit per clock cycle. It reads the source, elaborates the design, maps the register chain to flip-flops, and optimizes the logic for area and timing. This process efficiently generates the gate-level representation, helping designers verify functional and timing correctness while targeting specific platforms.

### APPLICATIONS:

An 8-bit shift register is used widely in digital systems for temporarily storing and shifting data bits in series. Its applications include data serialization and deserialization, timing adjustment, and data manipulation in communication systems. It helps reduce the number of I/O pins needed by allowing serial input and parallel output or vice versa. Shift registers are also used in digital delays, counters, and in creating finite state machines

### PROCEDURE TO CREATE A LAB:

#### **Step 1: Set up Project Directory**

1.1: Create a directory for your lab: `mkdir <project_name>`

1.2: Open lab directory: `cd <Project_name>`

1.3: Create directory for project: `mkdir shift_reg`

1.4: Open project directory: `cd shift_reg`

#### **Step 2: Create Design file**

2.1: Open gvim editor for design with command: `gvim shift_reg.v`

### RTL Design Code:

```
module shift_reg (  
    input wire clk,  
    input wire rst,  
    input wire [7:0] data_in,
```

```
output reg [7:0] q
);
always @(posedge clk or posedge rst) begin
    if (rst) begin
        q <= 8'b00000000;
    end else begin
        q <= {q[6:0], data_in[7]};
    end
end
endmodule
```

2.2: Save the file using command- :wq!

### Step 3: Synthesize the Design with Yosys

Invoke Yosys in a new terminal by the following command: yosys

```
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yosys -- Yosys Open SYnthesis Suite

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Yosys 0.9 (git sha1 1979e0b)

yosys> script shift_reg.ys
```

#### 3.1: Use the following command to Synthesize:

```
# Read the Verilog source file

read_verilog shift_reg.v

# Set the top-level module and check hierarchy

hierarchy -check -top shift_reg

# high level stuff
```



### OBSERVATION:

An 8-bit shift register is a sequential digital circuit that stores and shifts 8 bits of binary data either to the left or right with every clock pulse. It operates like a queue where data enters serially at one end and shifts through the register stages to the opposite end.

After this lab you will be able to observe generated synth.v file and the schematic representation of your design.

### OUTPUT:

Synthesized Netlist (shift\_reg\_synth.v)

### CONCLUSION:

The Yosys synthesis output confirms the correct implementation of the **8-bit Shift Register**, showing the RTL design was successfully mapped to a gate-level netlist. This netlist preserves the intended structure, with distinct combinational circuits for arithmetic and logic functions, all governed by a properly synthesized control logic.